

Name: Kyle Hankosky

50/100

CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, SPRING 2025, CLOSED BOOK, TIME: 60 MIN

The endothermic gas phase reaction $A \rightarrow B + C$ is carried out in a single steam-jacketed fluidized bed reactor, where the temperature of the steam circulating through the jacket remains constant at $T_a = 300^\circ\text{C}$.

For the following data, calculate the steady-state reactor temperature.

Conversion: $X = 85\%$

Heat-transfer area of the jacket: $A = 5 \text{ m}^2$

Overall heat transfer coefficient of the jacket: $U = 500 \frac{\text{J}}{\text{s.m}^2.\text{K}}$

Heat of reaction per unit mole of species A at 25°C : $\Delta_r H = 120 \frac{\text{kJ}}{\text{mol}}$

Feed temperature: $T_0 = 40^\circ\text{C}$

Feed molar flow rates: $F_{A0} = 10 \frac{\text{mol}}{\text{s}}$, $F_{I0} = 2 \frac{\text{mol}}{\text{s}}$ (inert), $F_{B0} = F_{C0} = 0$

Heat capacities (assume constant) $c_{pA} = 90 \frac{\text{J}}{\text{mol.K}}$, $c_{pB} = 60 \frac{\text{J}}{\text{mol.K}}$, $c_{pC} = 40 \frac{\text{J}}{\text{mol.K}}$, $c_{pI} = 50 \frac{\text{J}}{\text{mol.K}}$

Reactor energy balance: $\frac{dU}{dt} = \sum_{\alpha} F_{\alpha 0} h_{\alpha}(T_0) - \sum_{\alpha} F_{\alpha} h_{\alpha}(T) + \dot{Q} + \dot{W}_s$

Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$

$U \cdot A \cdot T_a = \dot{Q} = (500)(5)(300 - 277.15) =$
 $\dot{Q} = 1432875 \frac{\text{J}}{\text{s}}$

$\frac{dU}{dt} = 0$

	In	out
F_{A0}	10	1.5
F_{B0}	0	8.5
F_{C0}	0	8.5
F_{I0}	2	2

$h = \int_{T_{ref}}^{T_{act}} C_p dT$
 $h = C_p(T_{act} - T_{ref})$

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needs ΔH

$$0 = \sum E_{\text{in}} h_r(T_0) - \sum E_{\text{out}} h_r(T) + \dot{Q} - \Delta H(T) \Delta F_A? \quad \text{all energy in } \frac{J}{s} \checkmark$$

$$\sum E_{\text{in}} h_r(T_0) = 10(90)(40 + 273.15) + 0 + 0 + 2(50)(40 + 273.15)$$

281835 31315

$$\sum E_{\text{out}} h_r(T) = (1.5)(90)(T + 273.15) + (8.5)(60)(T + 273.15) + 8.5(40)(T + 273.15) + (2)(50)(T + 273.15)$$

$$\sum E_{\text{in}} h_r(T) = 313150 \quad \text{cancel } \frac{J}{s}$$

$$\sum E_{\text{out}} h_r(T) = (T + 273.15) [1.5(90) + 8.5(60) + 8.5(40) + 2(50)] = (T + 273.15)(1085)$$

$$0 = \underset{\text{In}}{313150} - \underset{\text{out}}{(T + 273.15)(1085)} + \underset{\dot{Q}}{\cancel{1432875}} - \underset{\Delta H}{\cancel{8.5(1000)(120)}}$$

$$\Delta h_{\text{I}} = 31315 - 100(T + 273.15)$$

$$\Delta h_{\text{A}} = 281835 - 135(T + 273.15) + 8.5(1000)(120)$$

$$\Delta h_{\text{O}} = 0 - 510(T + 273.15)$$

$$\Delta h_{\text{C}} = 0 - 340(T + 273.15)$$

$$\dot{Q} = \cancel{1432875} \text{ UAT} = (500)(5)((T + 273.15) - (300 + 273.15))$$

$$0 = 31315 - (T + 273.15)(1085) + 8.5(1000)(120) + 2500((T + 273.15) - 573.15)$$

$$T = -3.5^\circ \text{C}$$

